

Crucible Manufacturing Procedure

ForgeCore Industries — Production Department

Document ID: MFG-SOP-001 | Revision: 22 | Effective Date: 2025-03-01

1. Process Overview

ForgeCore crucibles are produced by either isostatic pressing (FC-45, FC-90) or slip-casting (FC-72) followed by controlled drying, high-temperature firing, and finishing operations. Total cycle time from raw material to finished goods is 12-18 days depending on product family and size.

2. Bill of Materials Summary

Product	Primary Material	Binder	Additives
FC-45	Silicon carbide 75-85%	Clay + lignin	Si powder 5%
FC-50	Graphite flake 80-90%	Pitch + resin	Boron carbide 3%
FC-72	Clay + graphite 60/40	Water + deflocculant	Grog 10%
FC-90	Zircon sand 80-90%	Clay + organic binder	Stabiliser 2%

3. Mixing Procedure

Raw materials are weighed per the batch recipe (typically 250-500 kg per batch) and loaded into a high-shear Eirich mixer. Dry mixing proceeds for 5 minutes, followed by gradual addition of binder solution over 3 minutes, then wet mixing for 12-15 minutes. Mix temperature must be maintained between 18-25°C throughout.

Critical control points:

- Mix homogeneity verified by torque curve analysis
- Moisture content must be within $\pm 0.5\%$ of target before discharge
- Discharge temperature must not exceed 35°C

4. Forming

FC-45 and FC-90 are formed by cold isostatic pressing at 35-45 MPa. The rubber mould is filled with batch material, sealed, and subjected to high-pressure water in a CIP chamber. Pressing cycle: 90 seconds hold at peak pressure, 30-second depressurisation.

FC-72 is formed by slip casting into plaster moulds. Slip viscosity is controlled at 800-1200 cP using a Brookfield viscometer at 100 rpm. Mould residence time is 6-8 hours for standard sizes, up to 14 hours for XL sizes. Cast pieces are trimmed after 4 hours of controlled drying.

5. Drying

Formed pieces enter controlled drying chambers with the following profile:

Stage 1 — 25°C, 80% RH, 24 hours (prevention of skin formation)

Stage 2 — 40°C, 60% RH, 48 hours (bulk moisture removal)

Stage 3 — 80°C, 30% RH, 24 hours (residual moisture)

Final moisture content target: <0.5%

6. Firing

Dried pieces are loaded into gas-fired shuttle kilns with carefully controlled temperature profiles. Maximum firing temperatures by product:

Product	Pre-heat	Body Rise	Peak	Hold	Cool
FC-45	RT→200°C in 6h	200→1300°C in 10h	1,350°C	2h	Natural to 200°C
FC-50	RT→300°C in 8h	300→1500°C in 12h	1,550°C	3h	Natural to 300°C
FC-72	RT→200°C in 5h	200→1100°C in 9h	1,150°C	2h	Natural to 150°C
FC-90	RT→200°C in 6h	200→1400°C in 11h	1,450°C	3h	Natural to 200°C

7. Finishing and Inspection

After firing, each crucible is visually inspected and dimensionally verified per QC-SOP-001. Surface defects >2 mm depth or >5 mm length are grounds for rejection. Acceptable pieces are sanded, marked with lot code and size designation, packaged in foam-lined crates, and moved to finished goods inventory.